

Chengtai Richard Li

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EDUCATION

University of California, Los Angeles (UCLA)

Los Angeles, CA

B.S. in Statistics and Data Science; Minors in Mathematics and Data Science Engineering

Expected June 2028

- **GPA:** 3.94/4.00 | **Awards:** UCLA Statistics Datathon 2025 – Award for Excellence in Analysis; Dean's Honor List
- **Relevant Coursework:** Data Structures and Algorithms, Machine Learning, Statistical Inference, Regression Analysis, Experimental Design, Optimization, Linear Algebra

EXPERIENCE

Full Stack Software Engineer

Mar 2025 – Present

Homie

Los Angeles, CA

- Build and ship core features for a cross-platform roommate management app using **React Native (Expo, TypeScript)** and a **Firebase** backend as one of **4** founding engineers on a **10-person** product team.
- Architected the Firestore data model and implemented **real-time multi-user sync** with TanStack Query, keeping chores, calendars, and shared lists consistent across devices for **20+** active users without manual refresh.
- Shipped a **barcode-based pantry scanner** to production by integrating Open Food Facts APIs covering **4M+** UPCs, Google Cloud Vision, and Firebase Cloud Functions leveraging **LLMs**, cutting grocery logging from **5** minutes of manual entry to under **30** seconds per grocery haul.
- **Tech:** React Native, Expo, TypeScript, Firebase, TanStack Query, OpenAI API, Google Vision, REST APIs, Git

Data Engineer Intern

May 2025 – Aug 2025

Royal Bank of Canada

Toronto, Canada

- Designed and trained an **Alternating Least Squares (ALS) matrix factorization** model for **collaborative filtering** in PySpark to recommend internal datasets to analysts on GRID, RBC's internal Tableau/Trino/Superset datahub, surfacing relevant tables from **50,000+** datasets for **500+** users.
- Productionized model outputs through an ETL pipeline that delivered daily recommendations into the GRID UI, reducing manual dataset search time by **80%**.
- Built and maintained **daily ETL pipelines** in Apache Airflow and PySpark that ingest, clean, and transform **3M+** rows across **5** internal data sources, implementing **data quality validation** so analysts had fresh, verified data each morning.
- **Tech:** Python, PySpark, SQL, Apache Airflow, ETL, CI/CD, Docker, Tableau, Machine Learning.

PROJECTS

Scrubs | *Swift, SwiftUI, Core ML, Apple Vision, Zetic*

Apr 2026

- **Won the Zetic Company Challenge** at LA Hacks 2026 (UCLA) out of **1000+** competitors.
- Architected a **privacy-first iOS app** that detects and redacts protected health information (PHI) – names, dates of birth, medical record numbers, and faces – from healthcare photos **entirely on-device**, eliminating HIPAA exposure risk.
- Engineered an **on-device computer vision pipeline** combining Apple Vision OCR for text extraction, a **Zetic** face detector, and a custom rule-based PHI classifier for entity recognition, validated across **1,000+** test photos with **zero network calls**.

Embers | *YOLOv11, Google Gemini, Next.js, Supabase, Flask, Python*

Apr 2025

- **Won Best Use of Google Gemini and 1st Place, Cold Hard Cache Track** at LAHacks 2025 (UCLA) out of 500+ competitors.
- Built an end-to-end **deep learning** video-to-inventory pipeline using YOLOv11 for **object detection** on a 2-minute home walkthrough, applying **multi-frame object deduplication and non-max suppression** to prevent double-counting across overlapping frames.
- Integrated **Google Gemini 2.5 for multimodal valuation**, passing cropped item snapshots through a reasoning prompt that infers brand, condition, and market value to generate per-item replacement estimates.

LEADERSHIP

Athletics Director, Association of Chinese Americans (ACA), UCLA

2025 – Present

- Lead event operations and budget coordination for a **1000+** member UCLA cultural student organization; built the board's internal reimbursement and stipend portal (**React, Supabase, Cloudflare Workers**), now used by **24** board members to process reimbursements.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, C++, R, Swift, HTML, CSS, Bash

Statistics & Modeling: Statistical Modeling, Regression Analysis, Hypothesis Testing, A/B Testing, Experimental Design, Feature Engineering

ML & AI: PyTorch, scikit-learn, NumPy, Pandas, OpenCV, Reinforcement Learning, Collaborative Filtering

Data: PySpark, Apache Airflow, ETL/ELT, Data Pipelines, Data Modeling, EDA, Data Visualization, Tableau, Jupyter

Frameworks & APIs: React, React Native, Next.js, Node.js, SwiftUI, FastAPI, Flask, Tailwind CSS, REST APIs

Cloud, DevOps & Databases: AWS, Firebase, Supabase, Docker, Kubernetes, Git, CI/CD, Agile/Scrum, Unit Testing, Redis, PostgreSQL, MySQL, Trino